



Consolidating Customer Data Across Legacy Systems for a Retail Chain

Project Objectives

- Apply concepts addressed from the course materials to a real-world scenario
- Analyze and justify decisions on data integration, storage, and migration in real-world data consolidation projects based on course content

Scenario and project

You will play the role of a junior data migration specialist at RetailCorp, a large retail chain operating over 200 stores across multiple regions. Over the years, the company has accumulated large amounts of customer data in various formats and across different systems.

After reviewing the project description, pipeline architecture diagram, and steps outline given below, you will make suggestions on its implementation based on the information you learned in this course.

Background

RetailCorp plans to unify all customer data into a modern, scalable data warehouse to facilitate accessible data analytics, personalized marketing, and improved customer experience. However, given the diversity of data formats, legacy systems, and the need to minimize disruption to ongoing operations, the company faces data integration, storage, and migration challenges.

They are in the planning stages, and their data architect has developed a consolidation pipeline flow diagram. As a data migration specialist, you will assist the architecture team in identifying plausible solutions for implementing the pipeline.

The senior data architect has supplied your team with

- Source systems descriptions
- A pipeline flowchart diagram
- Descriptions of each component in the pipeline
- Descriptions of the steps involved for integration, storage, and migration

Review these materials and then respond to the questions accordingly.

Source systems

The following describes the source systems for the migration:

- **E-commerce platform:** Customer data from online purchases, including transaction records, delivery addresses, and product preferences
- **Customer relationship management (CRM):** The system that stores business-to-business information, including names, contact details, and interaction history
- **Loyalty program database:** Stores information on customer loyalty points, rewards, and redemption history
- **Point of sales (POS):** Data from the transactions between the customer and RetailCorp, including the hardware and software used to complete the sales, record the transactions, and process payments
- **Marketing database:** Data produced through RetailCorp's marketing efforts, including campaign performance, demographics, website traffic, advertising efforts

RetailCorp consolidation project pipeline diagram

The data architect designed this flowchart pipeline diagram for RetailCorp's data consolidation project, which includes data integration, storage, and migration. Study the diagram and the component descriptions that follow.

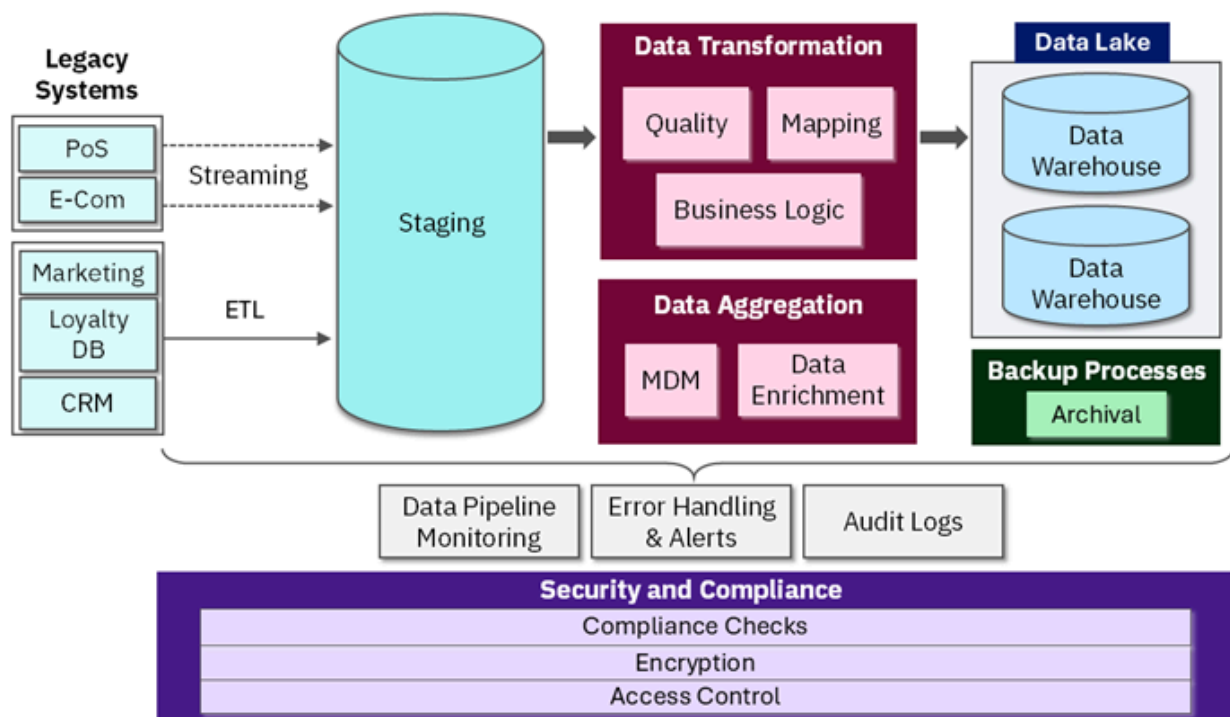


Diagram component details

1. Source Systems (Legacy Systems)

- CRM System
- Point of Sale (POS) System
- E-commerce Platform
- Marketing Database
- Loyalty Program Database

2. Data Ingestion Layer

- **Extractors:** Tools and scripts pulling data from the legacy systems

- **ETL (Extract, Transform, Load) Tool:** Centralized tool that completes the following tasks:
 1. **Extraction:** Includes tools and scripts pulling data from the legacy systems
 2. **Transformation:** Includes data cleansing and standardization
 3. **Load:** Placement of the data in the staging area
- **Streaming Data:** Processes streaming data to handle the real-time data ingestion from POS and E-commerce platforms
- 3. **Staging**
 - **Temporary Data Storage:** Temporary data storage before processing in a data lake
- 4. **Data Transformation Layer**
 - **Data Quality Services:** Tools for data cleansing, deduplication, and validation
 - **Data Mapping and Transformation:** Tools that map legacy system data structures to the new unified structure
 - **Business Logic Implementation:** Scripts or tools that implement business rules and logic into the data
- 5. **Data Aggregation Layer:** Combines the data from the staging area
 - **Master Data Management (MDM):** Merges the data into a single customer view
 - **Data Enrichment:** Addition of external data or additional information to enhance the customer records
- 6. **Data Storage**
 - **Data Warehouse:** The centralized repository for storing the consolidated data
 - **Data Lake:** Stores raw and semi-structured data
 - **Backup and Archiving:** Systems for data backup and long-term storage
- 7. **Monitoring and Logging**
 - **Data Pipeline Monitoring:** Tools to monitor the health and performance of the pipeline
 - **Error Handling and Alerts:** Mechanisms to log errors and provide alerts when issues occur
 - **Audit Logs:** Keeps track of data movement and transformations
- 8. **Security & Compliance**
 - **Data Encryption:** Both at rest and in transit
 - **Access Control:** Role-based access control to ensure only authorized personnel can access sensitive data
 - **Compliance Checks:** Ensures the pipeline adheres to industry regulations such as GDPR

Outline of the required steps for integration, storage, and migration

1. Data Integration

Data integration involves combining data from different sources into a cohesive system.

1.1. Assessing existing systems

Inventory of Data Sources: Identifying and analyzing all legacy source systems that store customer data

Data Mapping: Documenting the current data structures, schemas, and relationships within each source system and identifying key fields such as customer ID, name, contact information

Data Quality Assessment: Evaluating the quality of data in each system, checking for inconsistencies, duplicates, missing data, and outdated records

1.2. Defining the integration requirements

Standardizing the data: Establishing a common data format and schema to use across all systems, such as standardizing date and address formats

Identifying the integration points: Determining how the data will flow between systems into the central database by identifying the key data fields to synchronize

Selecting the appropriate integration tools: Choosing appropriate data integration tools and middleware, such as ETL tools like Talend and Informatica, to automate data extraction, transformation, and loading

1.3. Developing the data integration strategy

Extracting data: Defining the procedures for extracting data from legacy systems, which may involve using APIs, direct database connections, or flat file exports

Transforming data: Applying necessary transformations to align data with the target schema which may involve cleaning, deduplication, normalization, and data enrichment

Loading data: Implement processes for loading the transformed data into the central database, ensuring data integrity and consistency.

Synchronizing data: Establishing real-time and batch processing methods to keep the data synchronized across systems and the centralized data store

2. Data storage steps

The data storage strategy involves determining where and how to consolidate the data for storage.

2.1. Defining storage requirements

Planning capacity: Estimate the storage capacity required based on the volume of existing customer data and future growth projections.

Identifying performance needs: Assessing the performance requirements, including read/write speeds, query performance, and scalability

- Establishing compliance and security including industry standards, GDPR, and PCI-DSS
- Implementing robust security measures, including encryption and access controls

2.2. Choosing the storage solution

Selecting the centralized data store: Choosing a centralized store that supports your performance, scalability, and security needs, such as

Relational databases: Such as MySQL, PostgreSQL, or Oracle.

NoSQL databases: Such as MongoDB or Cassandra for more flexible data models

Data warehouses: Such as Amazon Redshift, Google BigQuery, or Snowflake for large-scale analytics

Choosing cloud or on-premises: Deciding whether to host the data storage on-premises, in the cloud, or a hybrid approach, considering factors like cost, scalability, and ease of management

2.3. Designing the data architecture

Designing the schema: Creating a unified schema that accommodates all customer data fields and relationships, ensuring it supports current and future use cases

Partitioning the data: Implementing data partitioning strategies to optimize performance and manage large datasets

Choosing data backup and recovery systems: Establishing a robust backup and disaster recovery plan, including automated backups and regular recovery tests

2.4. Implementation

Setting up the storage environment: Configuring the storage environment, setting up databases, user roles, access permissions, and security protocols

Data Loading and Indexing: Loading the initial batch of consolidated data into the storage system and creating the necessary indexes for query performance

3. Data Migration Steps

Data migration involves transferring data from the legacy systems to the new centralized storage.

3.1. Developing a migration plan

Defining the migration scope: Defining the scope of the migration, including which data sets will be moved, timelines, and any dependencies

Assessing risk: Identifying potential risks in the migration process, such as data loss, downtime, and performance impacts, and developing mitigation strategies

Cleansing the data: Cleaning the data to remove duplicates, correct errors, and ensure consistency before the migration

3.2. Choosing a migration approach

Consider

- **Big Bang Migration:** Moving all data in a single operation is faster but may involve more downtime
- **Phased Migration:** This method moves data in stages, allowing for gradual testing and adjustments with less risk but longer timelines.
- **Parallel Running:** Running old and new systems in parallel for a period to ensure the new system works as expected before fully switching over

3.3. Preparing the data for migration

Data Mapping: Mapping legacy data fields to the new schema, ensuring accurate data migration

Choosing migration tools: Selecting and configuring migration tools such as Talend, AWS Data Migration Service, or Informatica to facilitate data transfer

Developing data validation scripts: Creating scripts to validate data pre- and post-migration to ensure accuracy and completeness

3.4. Executing the data migration

Initial migration: Performing an initial migration, moving the bulk of the data to the new system

Incremental updating: Applying incremental updates if necessary to sync any changes that occurred after the initial migration

Final Validation: Running the data validation processes to ensure the integrity, accuracy, and completeness of the migrated data